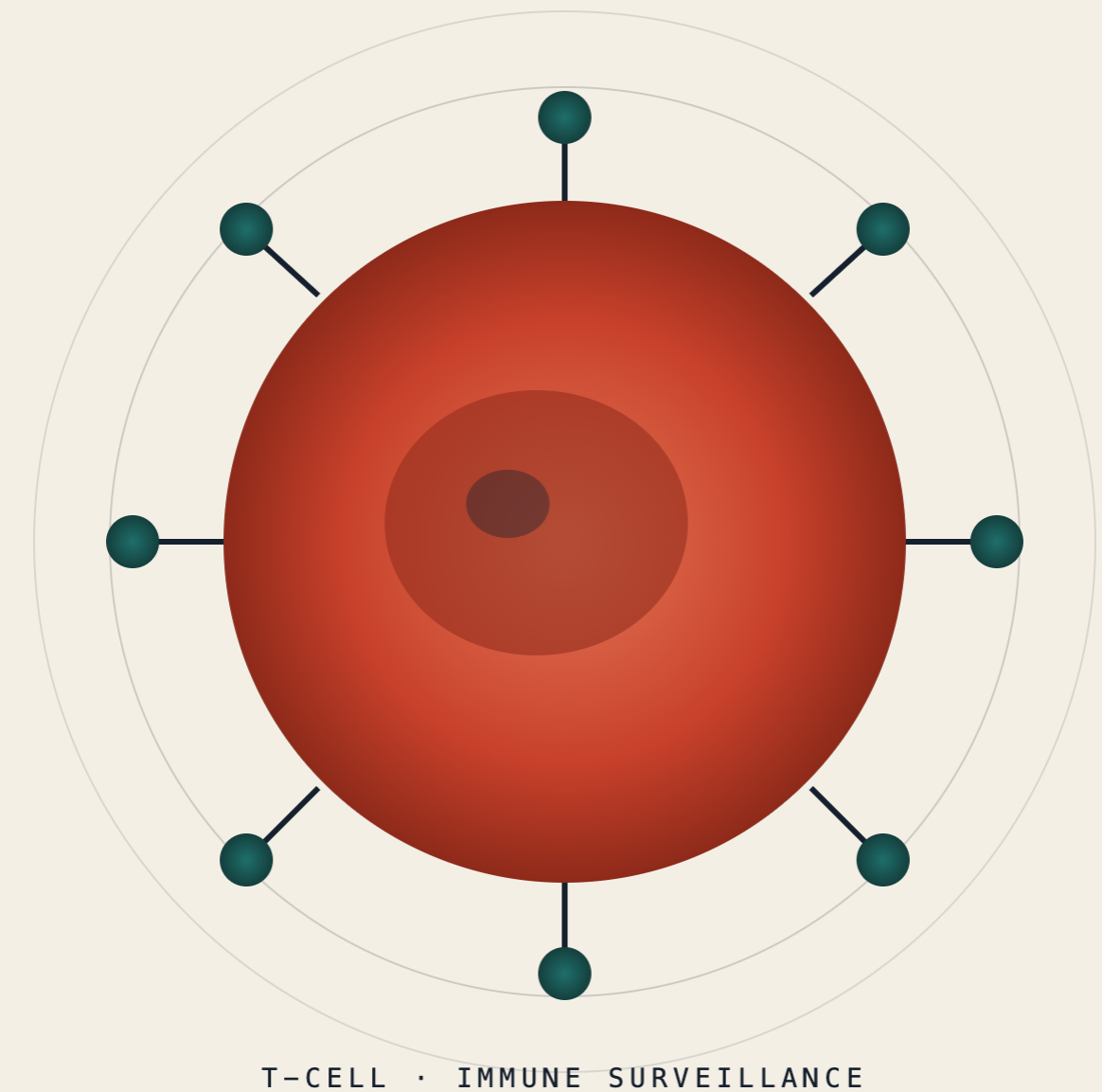


Immunity, *Limits*, and the Energy that Drives Them.

Course GSM · Friedrich-Schiller-Universität Jena

Lecture date 20 April 2026

Format 10-slide summary



Three systems, one body — what we'll cover today.

01

Immunity & Inflammation

How the body protects integrity through innate & adaptive defence — and how exercise tunes the inflammatory thermostat.

02

Limits of Performance

Definitions, determinants and the age-dependent ceiling on what muscular work the organism can deliver.

03

Energy Balance

From food to ATP — the energy systems that fuel contraction, and the macronutrients that feed them.

DEFENDING BODILY INTEGRITY

The immune system runs *two parallel programs* — one fast and unspecific, one slow and remembered.

FIRST LINE · INNATE

Innate Immunity

Non-specific, immediate response — no prior sensitisation required.

COMPONENTS	Skin, mucous membranes, neutrophils, macrophages, NK cells, complement
TRIGGER	Pathogen-associated molecular patterns (PAMPs)
SPEED	Activation within minutes

SECOND LINE · ADAPTIVE

Adaptive Immunity

Antigen-specific, slower to mobilise — but generates memory.

COMPONENTS	T lymphocytes (cellular), B lymphocytes (humoral), antibodies
TRIGGER	Specific antigens after sensitisation
SPEED	Days on first exposure, hours on re-exposure

A SELF-LIMITING CASCADE — WHEN IT WORKS.

Acute inflammation unfolds in *four phases* , each with a different cellular cast.

0 – 15 MIN



PHASE 1 · IMMEDIATE

Vascular alert

Mast cells & resident macrophages release histamine and cytokines; vascular permeability rises.

15 MIN – 2 H



PHASE 2 · EARLY

Neutrophil rush

Neutrophils dominate the acute response; complement amplifies phagocytosis and recruitment.

2 – 24 H



PHASE 3 · LATE

Macrophage cleanup

Macrophages clear debris, lymphocytes arrive, the adaptive response is initiated.

RESOLUTION



PHASE 4 · REPAIR

Restoration

IL-10, TGF- β suppress inflammation; tissue is repaired and homeostasis returns.

ACUTE · HOURS TO DAYS

Protective and self-limiting — fights infection, primes repair.

CHRONIC · WEEKS TO YEARS

Detrimental — implicated in CVD, T2D, autoimmunity, neurodegeneration, cancer.

SAME STIMULUS, TWO TIMESCALES

A single bout *mobilises* immune cells; a habit of training *lowers* the inflammatory baseline.

ACUTE BOUT

Mobilisation & *myokine release*

- **Lymphocyte & NK-cell trafficking** rises transiently in circulation.
- **IL-6 from muscle** acts as a myokine — anti-inflammatory signalling despite the spike.
- **Immune surveillance** enhanced as mobilised cells migrate to peripheral tissues.

CHRONIC TRAINING

Lower baseline, *better surveillance*

- Resting **IL-6, TNF- α , CRP** all reduced.
- T-cell numbers up; **vaccination antibody response** improved.
- Reduced infection risk in active people — but a **U-shape** appears at extreme endurance loads.

Regular moderate-to-vigorous activity enhances immune competence and dampens chronic inflammation — protecting against age-related decline and lifestyle disease.

PRACTICAL TAKEAWAY · §1 LECTURE NOTES

WHAT DO WE MEAN BY PHYSICAL PERFORMANCE?

Performance is defined by *four overlapping lenses* — biological, biochemical, lifespan and willed.

BIOLOGICAL-PHYSICAL

Muscular work per unit of time.

The sum of all determinants currently available to deliver mechanical body work.

$$P = \text{work} / \text{time}$$

LIFESPAN PERSPECTIVE

A trajectory, not a constant.

Genetically pre-set, behaviourally shaped — performance is part of biological, psychological and social development across the entire lifespan.

capacity(t) across the lifespan

BIOCHEMICAL ENERGY SUPPLY

ATP delivery per unit of time.

Aerobic and anaerobic energy provided by high-energy phosphates to fuel neuromuscular work.

aerobic + anaerobic ATP flux

EXTENDED DEFINITION

Value created against a goal.

Depends on biological, psychological and social resources, on internal and external stress — and on the *willingness to perform*.

resources × stress × will

WHAT SETS THE CEILING?

Two *primary determinants* shape capacity — and ten *modifiers* push it up or down.

Primary determinants

01

Biological structure

The anatomical scaffold — at the level of the whole organism and every sub-system within it.

02

Physiological functionality

How well the structure functions — across systems, control circuits and cellular metabolism.

Secondary determinants

- | | |
|-------------------------------|---|
| + Biological ageing processes | + Health as physiological functionality |
| + Disease & dysfunctionality | + Health-related lifestyles |
| + Nutrient supply | + Movement & physical activity |
| + Genetics & epigenetics | + Environmental factors |
| + Sensorimotor & perception | + Psyche & willingness to perform |

ENERGY SUPPLY BY DURATION

Four metabolic systems, each with its own *flow rate* and *time-to-fatigue*.

DURATION	ENERGY SOURCE	EXAMPLES	FLOW RATE · MMOL · KG ⁻¹ · S ⁻¹
< 2 s	Stored ATP	Power lift, javelin throw, golf swing, tennis serve	3.0 – 6.0
≤ 6–8 s	ATP + creatine phosphate (PCr)	Sprints, fast breaks, gymnastics routines	3.0 – 6.0
≤ 90 s	+ Anaerobic glycolysis (lactate)	200–400 m sprints, 100 m swim	1.5 – 3.0
~ 800 m	~50 % aerobic + ~50 % anaerobic	Middle-distance running	0.5 – 0.75
hours	Aerobic glycolysis → β-oxidation	Long-distance running, endurance events	0.24 – 0.40

"The faster you run, the sooner you get tired" — speed buys flow rate at the cost of capacity.

Capacities · ATP+PCr ≈ 19 mmol·g⁻¹ (≤8 s) · anaerobic glycolysis ≈ 50 mmol·g⁻¹ (≤60 s) · aerobic pathway limited only by substrate (hours – days).

IN = OUT — BUT ONLY ~40 % SURVIVES THE TRIP.

Macronutrients carry the energy; *ATP is the currency* the cell will accept.

80 · 10¹² cells

All require ATP to maintain structure and function.

< 40 %

Of food energy is available as utilisable / stored energy for skeletal muscle work.

689 kcal · ~34 % → ATP

Liberated by full oxidation of 1 mol glucose; about a third is conserved in ATP bonds.

FAT

9.3 kcal·g⁻¹

Largest store of biological energy. 50–70 % of fuel at low-to-moderate intensity; > 80 % in prolonged exercise.

1 triacylglycerol → **~460 ATP**

CARBOHYDRATE

4.1 kcal·g⁻¹

Primary fuel for anaerobic work and the CNS. Spares protein, primes fat catabolism. 60–70 % of intake for hard-training athletes.

1 glucose → **net 32 ATP**

PROTEIN

4.1 kcal·g⁻¹

Structural, not preferred fuel. RDA 0.83 g·kg⁻¹; athletes 1.2–1.8 g·kg⁻¹. Alanine cycle supplies ≤ 45 % of liver glucose in long efforts.

8 of 20 amino acids are **essential**

WHERE THE THREE THREADS MEET

Immunity, performance and energy are *one regulated system* — measured in ATP, modulated by movement.

01 · Immunity

Acute inflammation protects; chronic inflammation harms.

Regular activity tunes the inflammatory thermostat downward — IL-6 from muscle is a myokine, not just a stress signal.

02 · Performance

Capacity is biological structure × functional state × will.

Two primary determinants, ten secondary modifiers, and an age-dependent ceiling shape what the organism can deliver.

03 · Energy

Flow rate trades against capacity.

ATP-PCr pays out fast and runs dry; β-oxidation pays slowly but pays for hours. Intensity picks the system, duration empties it.

Next: *Cardiovascular & respiratory responses to exercise* — how the delivery infrastructure scales to meet the metabolic bill.